

Article

Evaluating Player Stress and Motivation Through Biofeedback-Controlled Dynamic Difficulty Adjustment

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Abstract

This study researches the effects of a biofeedback-based Dynamic Difficulty Adjustment (DDA) system on player stress management in a survival-horror video game. For this purpose, a game titled Code: Terror was developed, in which players' psychophysiological data—such as heart rate, stress level, and voice level—were collected in real time through a smartwatch. Based on these data, game parameters were dynamically adjusted. The adaptation process was guided by the “Relax-to-win” model, which aims to make the game easier as players remain calm, using their level of relaxation as a baseline. A total of 40 participants were recruited and divided into two reciprocal groups: one group experienced the game solely with the DDA system activated, while the other played first with the DDA system disabled and then with it enabled. Player experience was evaluated using the Intrinsic Motivation Inventory (IMI) and the Challenge Originating from Recent Gameplay Interaction Scale (CORGIS). Results showed that the dynamic biofeedback system significantly enhanced intrinsic motivation and perceived challenge, while also reducing physiological arousal, indicating its effectiveness in promoting emotional regulation—the inclusion of both gameplay conditions for participants allowed for the collection of more consistent and meaningful data. The results suggest that biofeedback-driven DDA systems have the potential to enrich player experience both emotionally and motivationally. Furthermore, the findings highlight the potential of such adaptive biofeedback mechanisms to serve as intelligent medical decision support tools for stress management, mental health monitoring, and personalized intervention strategies.

Keywords: biofeedback; dynamic difficulty adjustment; heart rate; horror; relax to win; video game



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1. Introduction

Contemporary video games have evolved beyond being mere tools of entertainment, transforming them into complex, experience-oriented systems that target players' psychological processes such as attention, emotion, and motivation [1]. This transformation has necessitated the development of dynamic and adaptable systems in game design, rather than relying on static structures. In this context, one of the most fundamental elements for ensuring a personalized and sustainable gaming experience is the ability to adjust the difficulty level in a manner appropriate to the player [2].

In traditional video games, players are typically offered the option to choose from preset difficulty levels. However, when these levels do not align with the player's actual skill level, the gaming experience may become monotonous or disengaging [3,4]. As a

solution to this issue, DDA has been developed as a technique that aims to automatically modify the game's difficulty in real time based on the player's performance [2].

The primary objective of DDA systems is to optimize the difficulty level based on each player's performance to sustain their engagement with the game [2]. By providing a level of challenge that aligns with the player's capabilities, these systems enhance both enjoyment and motivation [5–7]. Whereas fixed difficulty levels often fall short for players with varying skill and experience levels, DDA systems dynamically adjust the difficulty by analyzing in-game behavior and performance data. These systems can utilize a range of inputs, including in-game performance metrics [8–16], personality traits [17], hand movement [16], emotional states [13], psychological states, and physiological responses such as skin conductance [18–20], muscle contractions [21], heart rate [22,23], and heart sounds [19]. In doing so, DDA helps maintain the player in a state of “flow,” offering an experience that is neither too easy nor overwhelmingly difficult [4].

“Flow” refers to an intense state of motivation and enjoyment, as defined by Csikszentmihalyi, in which an individual is fully immersed in the activity, with diminished awareness of time and surroundings, and where the balance between personal abilities and the level of challenge is maintained [24].

DDA systems not only enhance the level of enjoyment but also increase player engagement and extend gameplay duration. Consequently, they have become an increasingly significant component in both commercial game development and in pedagogical or therapeutic game design contexts [25,26].

A notable example of a successful implementation of DDA can be found in Resident Evil 4 [27]. In this game, the difficulty level is automatically adjusted in the background based on the player's performance to maintain the player's “flow” state [24]. When players struggle—such as frequently dying, having low accuracy, or getting high damage—the number of enemies decreases, and the damage they inflict is reduced. More ammunition and health supplies become available in the environment. Conversely, when players perform well, enemies become more aggressive, deal greater damage, increase in number, and available resources become scarcer [28].

Conventional DDA systems typically rely solely on success/failure metrics and often overlook the player's emotional state. In recent years, there has been growing interest in integrating players' emotional conditions into DDA mechanisms. Accordingly, it has been argued that systems should be developed not only based on performance data but also on the player's psychophysiological state. This need has encouraged research on biofeedback-based DDA systems. In particular, the use of heart rate and facial expressions for adjusting game difficulty has been shown to influence the player experience positively [19,22,23,29,30].

In contrast to classical DDA systems, an experimental model known as the “Relax-to-win” dynamic difficulty strategy has emerged in recent years, reversing the traditional approach [23,31]. This system aims to base player success not solely on cognitive or motor skills but also on the ability to maintain emotional calmness. These systems operate by integrating with the game and monitoring the player's stress levels through physiological indicators such as heart rate, skin conductance, facial expressions, and voice level. The calmer the player remains, the greater their chance of in-game success. Conversely, when stress levels increase, the game becomes more difficult. This structure encourages a more mindful and self-awareness-oriented experience, particularly in horror and thriller genres. This approach represents a significant paradigm shift at the intersection of game design and psychology, as it emphasizes not only difficulty adjustment based on physical performance but also the relationship between emotional regulation strategies and game success. The “Relax-to-win” strategy enables the conceptualization of games as tools for stress manage-

ment and offers potential for integration with fields such as meditation, therapeutic game design, and cognitive-behavioral interventions [23].

Biofeedback-based systems aim to monitor players' physiological data in real time and dynamically adapt game content based on these inputs. Such systems enable a direct interaction between the player's emotional state and the game environment.

In the academic literature, affective games are defined as systems that shape the gaming experience based on biofeedback data [23]. Khaled and Yannakakis [32] implemented dynamic difficulty adjustment based on the player's stress levels. Biofeedback systems generally aim to enable users to perceive physiological data collected via sensors and to develop responses based on this information [33]. By allowing players' emotional states to be reflected within the game, these systems encourage the development of strategies for emotional regulation, resulting in heightened engagement and a greater sense of enjoyment. This, in turn, contributes to a more immersive and satisfying gameplay experience.

Emotion-based game adaptation systems encompass areas such as task structuring, personalization of difficulty settings, and the modulation of visual and auditory elements [33,34]. These systems may rely on either direct (e.g., muscle tension) or indirect (e.g., heart rate) forms of biological feedback data [23]. These technologies are also employed in relaxation training and the treatment of certain health conditions. They have been shown to produce outcomes comparable in effectiveness to conventional treatment methods in cases such as migraine [35,36] and chronic pain [37,38].

Evaluating player experience requires a multidimensional framework that goes beyond user interface or game mechanics, encompassing motivation, perceived difficulty, emotional engagement, and decision-making processes. Therefore, valid and reliable measurement instruments are essential. The IMI [39] and CORGIS [40] Scales used in this study provided a comprehensive means of evaluating player experience across both motivational and cognitive-emotional dimensions. The IMI scale enables the analysis of players' psychological engagement with the game, particularly through its subscales such as interest, perceived competence, effort, and pressure. The CORGIS scale contributes to the multidimensional assessment of DDA systems by measuring cognitive challenge, emotional load, performance demands, and decision-making processes.

This study aims to examine the effects of a biofeedback-based DDA system on perceived difficulty and intrinsic motivation in a survival-horror game context. To achieve this aim, we employed a "Relax-to-win" biofeedback-based dynamic difficulty adjustment approach, as it allows real-time personalization of gameplay based on the player's physiological and behavioral data. This method directly aligns with our goal of enhancing intrinsic motivation and optimizing perceived difficulty, as it adapts challenges according to the player's real-time psychophysiological state. For experimental data collection, a custom-designed game titled Code: Terror was developed. The system focuses on personalizing the gameplay experience in real time by monitoring the player's in-game physiological and behavioral data.

One of the original contributions of this study is the presentation of a highly applicable and innovative DDA model based on biological indicators such as heart rate, estimated stress level, and voice amplitude. This system, which can be seamlessly integrated into everyday gaming experiences, holds significant potential for enhancing player experience. Within the framework of the "Relax-to-win" approach, game difficulty is dynamically adjusted according to the player's real-time psychophysiological data. When stress levels rise, the system increases the difficulty; conversely, a state of physiological calmness leads to a more accessible experience. Preliminary findings suggest that this approach may be more effective than traditional performance-based DDA systems in improving player experience and enhancing intrinsic motivation.

2. Related Work

Hunicke [2] approached DDA systems within the context of “personalized game design” and identified them as a fundamental tool for enhancing player satisfaction. Similarly, Andrade et al [41] demonstrated that AI-supported difficulty adaptation positively influences the gaming experience when adjusted according to player performance.

Gilleade et al [42] emphasized that effective game design should simultaneously support the player, offer challenges, and deliver an emotional experience. They argued that games should become easier through supportive content when the player exhibits low performance, and conversely, increase in difficulty when the player performs well to prevent boredom. Moreover, eliciting emotional responses from the player was deemed crucial to the success of the game’s design.

Yannakakis and Hallam [43] revealed that evaluating player responses not only through behavioral data but also through psychophysiological indicators introduces a new dimension to game experience research. Such multidimensional assessment approaches play a critical role in accurately evaluating the effectiveness of DDA systems. In this context, purely quantitative data—such as in-game scores or completion times—are insufficient. The motivational and emotional bonds players establish with a game can only be meaningfully interpreted through valid and reliable instruments capable of multidimensional evaluation.

Robin Hunicke’s study [2] stands out as one of the pioneering works outlining both the philosophical and technical foundations of DDA systems. Hunicke argued that in order to sustain player experience, the difficulty level must be balanced to support the “flow” state [24]. The developed Hamlet prototype dynamically adapts in-game variables such as enemy density and resource distribution by analyzing player data in real time based on inventory theory and probabilistic models. Her findings challenged prevalent assumptions regarding common “difficulty complaints” in game design and showed that, when implemented correctly, DDA can enhance player engagement and support the flow state.

Mi and Gao [7] developed a novel DDA approach that focuses not only on player performance but also on the level of interaction and engagement. This system employed a machine learning model to estimate the player’s engagement level from gameplay data and adapted the game’s difficulty in real time. When engagement decreased, game content was restructured to become more engaging. Experimental results indicated that this approach resulted in higher user satisfaction and longer gameplay duration compared to static or performance-only based difficulty adjustments.

The approach developed by Fuchs et al. [12] aims to optimize a DDA system tailored to the player’s individual abilities using machine learning. In this two-phase system, an agent is first trained based on the player’s behavior, followed by the generation of a more challenging opponent. Tests conducted on the FightingICE platform revealed that these personalized opponents improved the player experience compared to conventional algorithms.

Duraisamy et al. [44] developed a system that supports player interaction and in-game adaptation using brain–computer interface (BCI) technologies. Players controlled the game solely through eye blinks, while their attention and mental fatigue levels were monitored via electroencephalography (EEG) signals. Based on these signals, the system adapted the game content in real time, improving both accessibility and the balance of cognitive load.

Orozco Mora et al. [45] developed a DDA system based on stress detection in a virtual reality (VR) environment. Participants’ stress levels were derived from electromyography (EMG) and heart rate data, which were processed using a machine learning model running on a Raspberry Pi. In-game variables such as enemy frequency and damage output were dynamically adapted to this data. It was found that damage-based adaptation resulted

in increased physical challenge and frustration, highlighting the critical importance of parameter selection in stress-based DDA systems.

Negini et al. [20] monitored players' emotional states using galvanic skin response (GSR) data in a first-person shooter (FPS) game and adapted in-game elements (NPC reactions, environmental features) in real time. When comparing explicit and implicit biofeedback conditions, higher levels of immersion were reported under explicit feedback. The system not only adjusted difficulty but also adapted the emotional atmosphere of the game, offering a more comprehensive experience.

Liu et al. [19] developed a real-time difficulty adaptation system based on players' anxiety levels. Anxiety was assessed using heart rate and skin conductance data; the game became easier under high anxiety and more difficult under low anxiety. While this approach did not explicitly refer to the "Relax-to-win" principle, it fundamentally aligns with its core philosophy. The findings indicated that this system had a more positive impact on player experience compared to traditional performance-based DDA.

Andrew et al. [46] implemented a DDA system by monitoring players' real-time facial expressions using a face and emotion recognition framework. Emotional reactions such as fear and anxiety were analyzed to adjust the game's difficulty based on the player's emotional state dynamically. When highly stressed expressions were detected, difficulty was decreased; conversely, when a relaxed state was sensed, the challenge increased. Thus, the system did not simply aim to relax the player for guaranteed success but instead offered a balanced adaptation model.

Nogueira et al. [23] developed a biofeedback-based DDA system using electrodermal activity (EDA) data in the procedural horror game *Vanish*. Game difficulty was dynamically adjusted based on players' stress levels, and the system personalized the experience by balancing both states of stress and relaxation.

In a study by Moschovitis and Denisova [22], a DDA system using heart rate-based biofeedback was implemented in a horror-themed game called *Caroline*. Players' heart rate levels were monitored in real time, and game difficulty was automatically increased or decreased accordingly. The study adopted the "Relax-to-win" principle, adjusting difficulty to encourage players to remain calm. During the experimental process, dimensions such as intrinsic motivation, cognitive load, emotional responses, performance, and decision-making were measured. While no statistically significant change was observed in decision-making difficulty, the implemented DDA system was found to significantly enhance players' intrinsic motivation. However, no significant effects were reported in cognitive difficulty, emotional challenge, or performance-related dimensions. This study is considered one of the notable examples in which biofeedback-based DDA systems were tested within a real gameplay environment.

In Table 1, selected examples from the related literature are summarized and presented in comparison with this study.

Table 1. Related studies.

Reference	Biofeedback Input	Genre	Description
Moschovitis and Denisova [22]	HR	Horror	In this study, a biofeedback-based DDA system was implemented, adopting the "Relax-to-win" principle. Unlike prior research that relied solely on heart rate data, our study utilized a combination of heart rate, sound level, and heart rate variability. To enhance accessibility and practicality, data were collected exclusively through a smartwatch, avoiding the need for additional sensors. Although the same measurement scales were employed, previous studies yielded significant results only on the IMI scale, whereas our study found significant results on both the IMI and CORGIS scales. While the test games in both studies are similar in terms of their core genre, our research distinguished itself by selecting the survival-horror genre instead of general horror. While horror games typically focus on elements of fear and suspense, the survival-horror genre incorporates limited resources, a pervasive sense of vulnerability, and the struggle for survival, rendering the experience of fear more profound and personal.

Table 1. Cont.

Reference	Biofeedback Input	Genre	Description
Nogueira et al. [23]	HR, SC, EMG	Horror	In this study, biofeedback-based DDA was used, and the relax-to-win principle was adopted. In both studies, heart rate data is common, and other inputs are different. As for the test environment, there is a horror/survival-horror difference as in the previous example.
Andrew et al. [46]	-	Survival-horror	In this study, instead of biofeedback-based DDA, the principle of DDA with facial expression monitoring was adopted. The difficulty was decreased when the perceived player expression was stressed and increased in the opposite case. The system does not work on the relax-to-win principle.
Liu et al. [19]	PPG, ECG, EMG, EDA	Sports	In this study, biofeedback-based DDA was used. Although the relax-to-win principle is not directly mentioned, there are some parallel cases. Many different inputs were used as biofeedback inputs, but, as in other studies, they are far from accessible and practical. As a game type in the test environment, a completely different type of sport was preferred.
Negini et al. [20]	GSR	FPS	A non-biofeedback-based DDA system was used. One of the scales used is IMI, which is in common with our study. As it is not biofeedback-based, there is no relax-to-win principle. A first-person shooter game was preferred as the test environment.
Orozco Mora et al. [45]	ECG, EMG	FPS (VR)	In this study, biofeedback-based dynamic difficulty adjustment was used. Relax-to-win principle is not used in this study. The study was developed as a VR game using ECG and EMG sensors. First-person shooter was preferred as the game genre in the test environment.
Duraisamy et al. [44]	EEG	Platformer	This study uses biofeedback-based dynamic difficulty adjustment. The relax-to-win principle is not used in the study. The study is about controlling and playing the game with EEG sensors. A platformer game was chosen as the test environment.
Fisher et al. [13]	EEG	FPS	This study uses biofeedback-based dynamic difficulty adjustment. The relax-to-win principle is not used in the study. Rather than employing a single, effective DDA method, the study utilized techniques such as performance-oriented, emotion-oriented, or hybrid approaches. An FPS endless survival genre was chosen as the test environment.
García-Ramón et al. [16]	LMC	Serious	This study investigates player engagement in hand rehabilitation games by analyzing EEG signals under different difficulty adjustment strategies. Hand movements, tracked using a Leap Motion Controller (LMC), served as input for dynamic difficulty adjustment (DDA). Specifically, the study compares a customized dynamic difficulty mode in which game challenges are adapted according to the player's range of motion and performance with a random difficulty mode, where challenges are assigned without personalization.
Fuchs et al. [12]	-	Fighting	No biofeedback or similar systems are used in the study. A study was conducted on the use of Reinforcement learning and DDA. A fighting game was preferred as a test environment.
Li et al. [15]	-	FPS	This study investigates the optimization of player experience in FPS games through the dynamic adjustment of in-game audio cues based on player performance. A Gaussian Process Regression (GPR)-based DDA mechanism is employed to analyze performance metrics, subsequently modulating the volume of enemy sounds and other audio cues.
Romero-Mendez et al. [14]	-	Fixed Shooter	This study uses a deep learning (DL) approach to enhance gameplay by dynamically adjusting game difficulty based on a player's skill level. A fixed shooter game was preferred as a test environment.
Mi and Gao [7]	-	-	No biofeedback or similar systems are used in the study. It is a study conducted to gain a new perspective on the differentiation of the DDA system. A method called EDDA (Engagement-oriented Dynamic Difficulty Adjustment) was developed.
Hunicke [2]	-	FPS	It is the work that defines the center of the literature on DDA systems. It describes the theoretical and practical principles of DDA.

Contributions to the Literature

This study offers an original contribution to the literature by examining the effects of a biofeedback-based DDA system on player experience within a custom-developed, high-production-quality survival-horror video game featuring advanced graphical realism. The research introduces a novel approach that reduces hardware dependency by implementing DDA using data collected solely via a smartwatch, thereby advancing the development of more user-friendly systems. Unlike the multi-sensor systems frequently employed in the literature, this study collected data on heart rate, voice level, and heart rate variability-based stress exclusively through a smartwatch. As a result, the system was simplified from a hardware perspective while becoming more practical and accessible. The ability to obtain real-time physiological data without the need for additional hardware facilitates the integration of DDA systems into broader user groups, offering cost-effective and portable solutions in the field of

gaming technologies. In this respect, the study addresses a significant gap in the literature, contributing to both methodological innovation and practical application.

The collected biofeedback data were transmitted to the game system in real time via a local network, and the game difficulty was dynamically adjusted in accordance with the “Relax-to-Win” principle. Unlike previous methods, this approach assessed the player’s physiological and emotional states multidimensionally, demonstrating the effectiveness of the DDA system in a real gaming environment. The research findings indicate that participants who played with the DDA system scored significantly higher on both the IMI and CORGIS scales and exhibited lower heart rate ranges, consistent with the “Relax-to-Win” approach.

Results from the IMI and CORGIS scales revealed a significant relationship between player experience and intrinsic motivation. These instruments have also been successfully applied in previous biofeedback-based DDA studies framed by the “Relax-to-Win” approach. By applying these measurement tools for the first time within a survival-horror game context, this study supports their validity in such settings. Thus, the research contributes a genre-specific perspective to the relationship between motivational processes and game experience, extending existing applications.

Finally, the game used in this study was specifically designed within the survival-horror genre, marking one of the first large-scale experimental explorations of the genre’s potential for DDA systems. The survival-horror genre offers a multi-layered emotional and cognitive experience through elements such as feelings of isolation, limited resources, and constant threat. While horror-themed games typically emphasize atmospheric tension and sudden scares, survival-horror deepens this experience with dynamics of vulnerability, resource management, and survival, adding a more personal dimension. In this context, the genre aligns strongly with the “Relax-to-Win” approach, which rewards stress management and emotional balance. Consequently, this study provides an original and significant contribution to the literature as one of the first experimental examples demonstrating the applicability of a comprehensive DDA system within the survival-horror genre.

3. Experimental Method

This research presents an innovative contribution to the literature by implementing a DDA system that collects physiological data solely via a smartwatch, thereby offering both hardware simplification and enhanced accessibility. Furthermore, the significant findings obtained through the IMI and CORGIS scales contribute to the literature by holistically addressing the relationship between motivation and player experience within an applied context. The choice of the survival-horror genre for the experimental process aligns closely with the “Relax-to-Win” principle by emphasizing players’ stress management abilities, thereby advancing the integration of this genre with comprehensive DDA systems.

Traditional dynamic difficulty systems adjust the game’s difficulty level based on the player’s in-game performance. However, in this study, difficulty adjustments were made directly based on psychophysiological parameters, independent of player achievement levels [47]. In the proposed system, as the player’s tension level increased, the game experience became more challenging.

The system’s applicability was tested using a specially developed survival-horror video game. A total of 40 participants were randomly assigned into two groups within the study. The first group played the game with dynamic difficulty enabled and then completed the IMI and CORGIS questionnaires. The second group initially experienced the game under a fixed difficulty setting, after which survey data were collected. This group then replayed the game with dynamic difficulty activated and completed the surveys a second time.

The experimental designs employed to test the hypotheses are presented in Figures 1 and 2. The primary research hypotheses developed for this study are as follows:

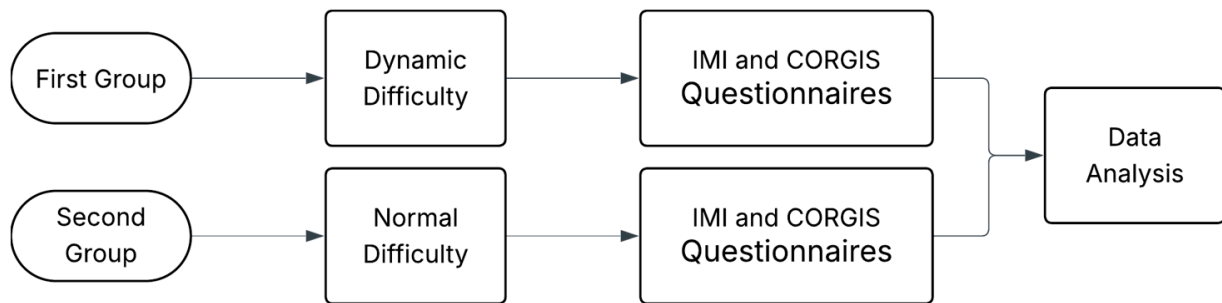


Figure 1. Experimental Design I.

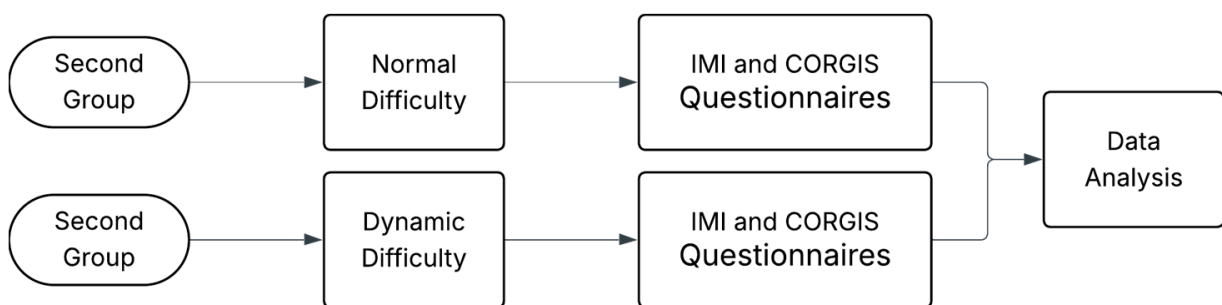


Figure 2. Experimental Design II.

Experimental Design I:

H1₁: *There is a significant difference in IMI and CORGIS scores between the first group (dynamic difficulty) and the second group (fixed difficulty).*

H0₁: *There is no significant difference in IMI and CORGIS scores between the first group (dynamic difficulty) and the second group (fixed difficulty).*

Experimental Design II:

H1₂: *There is a significant difference in IMI and CORGIS scores within the second group between the fixed and dynamic difficulty modes.*

H0₂: *There is no significant difference in IMI and CORGIS scores within the second group between the fixed and dynamic difficulty modes.*

H1₃: *Experimental Design II demonstrates a higher effect size compared to Experimental Design I.*

H0₃: *Experimental Design II does not demonstrate a higher effect size compared to Experimental Design I.*

H1₄: *In sessions where dynamic difficulty is applied, participants' heart rate ranges are significantly lower compared to sessions with fixed difficulty.*

H0₄: *In sessions where dynamic difficulty is applied, participants' heart rate ranges are not significantly lower compared to sessions with fixed difficulty.*

The research was conducted within an experimental design framework, where participants engaged in two different gaming scenarios: one with the biofeedback-based DDA system activated and one without. By comparing these experimental conditions, the effects of the system on player experience were analyzed.

The independent variable in the study was the presence or absence of the biofeedback-based DDA system in the game (present/absent). The dependent variables were defined as players' intrinsic motivation levels and perceived game difficulty. Post-experiment, data concerning participants' experiences were collected via validated scales and structured survey forms. These data were subjected to statistical analysis to evaluate the effectiveness of the proposed DDA system.

3.1. The Game

As part of this study, an original survival-horror video game titled *Code: Terror* was developed to experimentally test the proposed hypotheses. The game was designed using the Unity 6 (Unity 6000.0.40f1, Unity Technologies, San Francisco, CA, USA) game engine. The narrative unfolds around a character stranded in an isolated forest due to a vehicle breakdown, who encounters an abandoned mansion deep within the woods, where unusual events unfold. While uncovering the secrets of this mysterious structure, the player must also struggle to survive.

The mansion is depicted as a former research facility where illegal biological experiments were conducted. As a consequence of these experiments, a zombie outbreak has emerged, posing a threat both within and around the mansion. The player is tasked with uncovering clues about the origin of the outbreak while defending against various threats. The game features hidden rooms, locked safes, puzzles, and locked doors. Initially, players only have access to melee weapons such as knives and axes, but as the game progresses, they can acquire firearms. The ultimate goal is to discover the key to a mysterious door located in the mansion's basement, thereby uncovering the core secrets of the narrative.

The game design aims to foster empathy with the character and enhance the sense of realism. Although originally planned as a third-person perspective experience, the game was restructured to a first-person perspective to amplify immersion (Figures 3a,b and 4). To maximize the effectiveness of the DDA system and cater to diverse player profiles, the game content is structured with clear guidance and informative elements. Tasks provide explicit instructions on what players need to do and which objects to interact with, while the interface offers constant hints and shortcut key information.

The game is designed as a semi-open world, providing players with multiple pathways for progression. Tasks can be completed non-linearly, and some areas are optional. This design grants players the freedom to shape their experiences and enhances replayability.

The atmospheric design was meticulously crafted to reflect the core elements of the survival-horror genre. Selecting an abandoned forest (Figure 3b) and a mansion (Figures 3a and 4) as settings reinforces feelings of isolation and threat. Atmospheric tension is heightened through dynamic lighting and shadow effects, narrow and dark corridors, limited visibility, and restricted light sources. Zombies do not appear frequently; instead, tension is maintained via environmental cues such as footsteps or an ambient sense of looming threat. Environmental sounds are dynamically modulated: natural sounds dominate forested areas, echoing interiors in the mansion, and oppressive low-frequency sounds in the basement.

To enhance the impact of the biofeedback-controlled DDA system, the player's heart rate is analyzed in real time and synchronized with an in-game heartbeat sound (Figure 5). This sound dynamically adjusts with the player's stress levels—sound level increases as the stress increases—thereby intensifying sensory tension and allowing players to experience

their psychological state more tangibly. Stress was estimated based on heart rate (HR) and its related heart rate variability (HRV), with further details of the calculation provided later in the manuscript.



(a)



(b)

Figure 3. Screenshots from the game: (a) an image from the cellar section, (b) a scene from the forest at the game's beginning.



Figure 4. In-game scene from the mansion, which serves as the primary setting for puzzles and encounters with zombies.

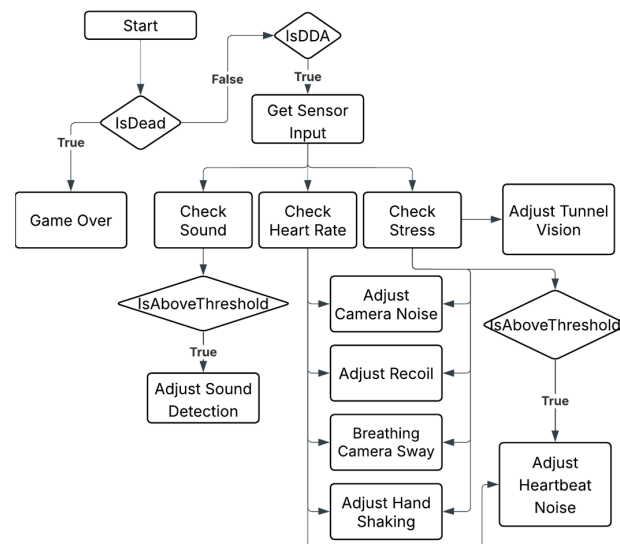


Figure 5. Player flowchart.

Enemies are designed in two types: standard zombies and a singular, special mutant zombie. Standard zombies pose a low-level threat, whereas the mutant zombie is more aggressive, resilient, and offers a higher difficulty challenge. The enemies' AI systems are dynamically configured (Figure 6). With dynamic difficulty enabled, zombies detect the sounds made by players and respond accordingly. Upon detecting the player, zombies enter a “chase” mode. If the player escapes the line of sight and hides for a sufficient time, the enemies pause at the last known location before returning to predefined patrol points on the map, ensuring a degree of randomness in their movement patterns.

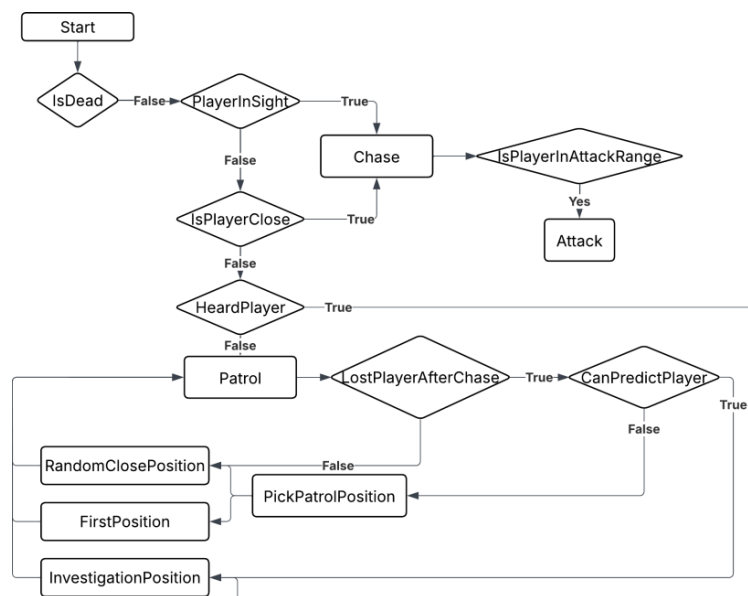


Figure 6. Flowchart of enemy behavior.

To complete the game, players must locate the key to a locked room in the mansion's basement. The game concludes when players successfully accomplish the tasks and survive. The total duration varies based on play style: experienced, task-focused players can complete the game in approximately 15 min, while exploratory or less experienced players may take up to 30 min. This variability demonstrates that the game experience is shaped by individual preferences and interaction levels.

In conclusion, Code: Terror is an original research tool developed to measure the effects of biofeedback-based dynamic difficulty adjustment on player experience. It is supported by rich narrative elements, atmospheric depth, and technical components, making it a comprehensive and innovative contribution to the field.

3.2. Dynamic Difficulty Adjustment

In this study, the implemented DDA system utilizes players' physiological data—heart rate, sound level, and stress levels—as biofeedback inputs, dynamically adapting the gaming experience based on these metrics. These physiological data were collected via smartwatch and transmitted to the game in real time through a custom-developed Wear OS (Wear OS 5.0, Google LLC, Mountain View, CA, USA) application over a local area network (LAN) connection.

Heart rate data was selected as the primary biological input due to its sensitivity to real-time fluctuations and its direct reflection of stress responses. Audio data captures both voluntary and involuntary vocal reactions from the player (such as shouting, deep breathing, or sudden increases in voice amplitude), enabling AI-controlled enemies to react to these sounds. This creates a more realistic mechanism for enemy detection, establishing a dynamic link between tension and interaction through sound.

Stress levels are primarily calculated via time-series analysis of heart rate data, which is then linked to various game mechanics such as hand tremors, screen darkening, and visual blurring. This calculation operates using a methodology similar to stress detection algorithms commonly used in smart devices. Heart rate (HR) and heart rate variability (HRV) metrics were used to estimate the user's stress level. The HR metric represents the instantaneous heart rate in beats per minute as measured directly by the wearable device. Since the Xiaomi Watch 2 (Xiaomi Inc., Beijing, China) lacks a dedicated HRV sensor, a PPG(Photoplethysmography)-based HRV metric was computed using a root mean square of successive differences (RMSSD). RR intervals were derived from the last 10 HR measurements, and the square root of the mean of the squared differences between successive RR intervals was calculated to provide a PPG-based HRV.

The sampling frequency of the heart rate measurement is approximately 10 Hz, based on the sensor's fastest available setting. A sliding window of 10 consecutive HR measurements was employed for the PPG-based HRV calculation and subsequent stress score computation, providing a 1 s temporal resolution for HRV-based assessments.

In this implementation, HR is derived directly from raw sensor readings without any smoothing or baseline normalization. The values are used as-is, reflecting immediate measurements, and no filtering or adjustment is applied to reduce noise or account for individual baseline differences. A weighted combination of HR and PPG-based HRV (each with a 0.5 contribution) was then used to calculate the stress score, which was constrained to a 0–100 scale. It should be noted that the normalization applies only for the stress calculation; the raw HR values are transmitted directly over the network without modification.

The average latency from the sensor reading to the game's feedback response is approximately 950 ms. This calculation represents the time elapsed from the initial moment the data is read by the sensor to the point it is transmitted and subsequently reflected back to the player as feedback.

In the physiological data obtained from the smartwatch, no anomalous or extreme values were detected during preprocessing. All heart rate values fell within physiologically plausible ranges, and thus no data were removed or adjusted. Consequently, additional procedures for outlier handling were not required.

The audio data were recorded at 44.1 kHz in mono PCM format and processed in real time using root mean square (RMS) calculations to estimate sound intensity. The RMS values were then converted into decibels (dB) to normalize the measurements. This procedure inherently filtered out implausible spikes or noise artifacts, and no extreme values requiring further adjustment were observed.

Within the game, these data influence 16 distinct parameters directly, which in turn indirectly affect other secondary game variables. The directly impacted parameters are hand tremor speed, hand tremor amplitude, weapon recoil intensity, breathing rate, breathing intensity, screen shake speed, screen shake amplitude, vignette (darkening effect) intensity, vignette speed, vignette position, vignette fade, heartbeat sound volume, enemy movement speed, enemy field of view, enemy vision distance, and enemy detection range for player.

To quantitatively describe game difficulty, the variability of in-game parameters under dynamic difficulty and the fixed parameters corresponding to normal difficulty are presented in terms of their minimum and maximum possible values. In the normal difficulty condition, parameters are fixed at their minimum values, whereas in the dynamic difficulty condition, parameters vary within this range according to the governing formulas, with the upper limits denoted as the maximum values. Taking one of the main parameters as an example, the hand tremor speed may increase by up to five times under dynamic difficulty, thereby amplifying the player's motor challenge. As another example, the screen shake speed parameter can increase by up to ten times, intensifying the visual perturbations experienced during gameplay.

$$HR = \text{Heart Rate}$$

$$ST = \text{Stress}$$

$$SL = \text{Sound Level}$$

$$\text{shakeSpeed} = \frac{HR}{10} * ST * \text{baseShakeSpeed}, \quad \text{shakeSpeed} \in [0, 5] \quad (1)$$

$$\text{handAmplitude} = \text{baseAmplitude} * ST * 2, \quad \text{handAmplitude} \in [0.01, 20] \quad (2)$$

$$\text{recoil} = \text{baseRecoil} * ST * \text{recoilMultiplier}, \quad \text{recoil} \in [2, 1000] \quad (3)$$

$$\text{breathRate} = \frac{HR}{100}, \quad \text{breathRate} \in [0.5, 3.6] \quad (4)$$

$$\text{breathIntensity} = HR * \text{breathMultiplier}, \quad \text{breathIntensity} \in [0.05, 0.18] \quad (5)$$

$$N = \text{noiseSpeedMultiplier}$$

$$\text{noiseSpeed} = \min\left(5, \max\left(0, \frac{HR}{10}\right) * ST * N\right), \quad \text{noiseSpeed} \in [0.5, 5] \quad (6)$$

$$\text{noiseAmp} = ST * \text{noiseAmplitudeMultiplier}, \quad \text{noiseAmp} \in [0.01, 10] \quad (7)$$

$$\text{vignetteStrength} = \min(1, \max(0, ST)), \quad \text{vignetteSpeed} \in [0, 1] \quad (8)$$

$$\text{Lerp}(a, b, t) = (1 - t) * a + t * b$$

$$\text{vignetteSpeed} = \text{Lerp}(0.1, 3, ST), \quad \text{vignetteSpeed} \in [0.1, 3] \quad (9)$$

$$\text{vignettePosition} = \text{Lerp}(-0.2, 0.2, ST), \quad \text{vignettePosition} \in [-0.2, 0.2] \quad (10)$$

$$\text{vignetteFade} = \min(1, \max(0, ST)), \quad \text{vignetteFade} \in [0, 1] \quad (11)$$

$$\text{heartBeatVolume} = ST * \text{baseVolume}, \quad \text{heartBeatVolume} \in [0, 1] \quad (12)$$

$$\text{enemySpeed} = \text{baseSpeed} * \left(1 + \frac{ST}{20}\right), \quad \text{enemySpeed} \in [3, 18] \quad (13)$$

$$\text{enemyFOV} = \text{defaultFOV} + SL, \quad \text{enemyFOV} \in [110, 180] \quad (14)$$

$$enemySight = defaultSight + \frac{SL}{5}, \quad enemySight \in [15, 30] \quad (15)$$

$$y = enemyClosePlayerDetection$$

$$e = enemySightDistance$$

$$d = defaultPlayerDetection$$

$$y = \begin{cases} \frac{e}{2} & s > 35 \\ d & s \leq 35 \end{cases} \quad y \in [3.5, 30] \quad (16)$$

One of the most distinctive features of the DDA system is the direct translation of the player's physiological data into in-game feedback. When a certain stress threshold is exceeded, visual and auditory cues such as hand tremors, camera shakes, or vignette effects are activated on the player's screen. Additionally, as the stress level rises, a heartbeat sound begins to play within the game environment, with its volume proportionally adjusting based on the player's actual heart rate. This implementation aims to enhance realism and intensity by integrating the player's psychophysiological state into the game atmosphere.

The system was designed to be compatible with other smartwatches equipped with the necessary sensors and running the Wear OS operating system. When the gaming PC and smartwatch are connected to the same local network, synchronization is achieved via the watch's IP address, and data transmission is handled through the TCP protocol. This architecture offers a significant advantage in terms of scalability and accessibility for a portable and low-cost biofeedback-based DDA system.

3.3. Questionnaires

Upon completion of the game, player experience, as well as perceptions of difficulty and motivation, were assessed using validated survey instruments. For this purpose, the IMI and CORGIS scales were utilized. Cronbach's alpha coefficients indicating the internal consistency of both scales are presented in Table 2.

Table 2. Cronbach's Alpha Coefficients for CORGIS and IMI Scales Reliability.

	Cronbach's α	Mean $\pm$ Standard Deviation	
		Second Group (Normal)	First Group (Dynamic)
IMI	Interest/Enjoyment	34.70 $\pm$ 6.053	40.25 $\pm$ 7.383
	Perceived competence	24.95 $\pm$ 10.241	24.65 $\pm$ 12.966
	Effort/Importance	22.75 $\pm$ 4.024	27.35 $\pm$ 4.826
	Pressure/Tension	26.95 $\pm$ 5.195	30.70 $\pm$ 4.001
	Total: Intrinsic Motivation	109.35 $\pm$ 14.886	122.95 $\pm$ 16.916
CORGIS	Cognitive Challenge	50.10 $\pm$ 11.611	58.45 $\pm$ 10.354
	Emotional Challenge	19.80 $\pm$ 6.444	23.80 $\pm$ 3.650
	Performative Challenge	21.65 $\pm$ 3.391	24.10 $\pm$ 5.159
	Decision-Making Challenge	9.40 $\pm$ 4.284	12.40 $\pm$ 5.123
	Total: Perceived Challenge	100.95 $\pm$ 15.879	118.75 $\pm$ 16.814

The Intrinsic Motivation Inventory (IMI) [39,48] was employed to evaluate participants' intrinsic motivation levels within the video game context [22,49,50]. This instrument comprises 45 items scored on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree). IMI includes four subscales:

- Interest/Enjoyment—7 items;
- Perceived Competence—6 items;
- Effort/Importance—5 items;

Pressure/Tension—5 items.

Separate scores were calculated for each subscale, and the total IMI score was also used in statistical analyses.

Additionally, to assess players' perceived difficulty in the game, the 30-item CORGIS [40] questionnaire was administered, also using a 7-point Likert scale. CORGIS consists of four subscales:

Cognitive Challenge—11 items;

Emotional Challenge—9 items;

Performative Challenge—5 items;

Decision-Making Challenge—5 items.

Separate scores were computed for each subscale, along with a total CORGIS score.

The reliability of the CORGIS scale was evaluated using Cronbach's alpha. The values obtained for the subscales were as follows: cognitive challenge (0.89), emotional challenge (0.84), performative challenge (0.94), and decision-making challenge (0.86). To test the factor structure of the scale, both Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) were conducted [40].

Similar to CORGIS, the IMI scale is a reliable and valid instrument, with well-defined subscales and factor structures that have been tested in numerous studies [51].

Table 2 presents Cronbach's alpha values, means, and standard deviations for the CORGIS and IMI scales. For the IMI subscales, the Cronbach's alpha coefficients were as follows: Interest/Enjoyment (0.90), Perceived Competence (0.963), Effort/Importance (0.819), Pressure/Tension (0.881), and Total IMI (0.854), indicating sufficient internal consistency [52]. For the CORGIS scale, the coefficients were Cognitive Challenge (0.973), Emotional Challenge (0.672), Performative Challenge (0.803), Decision-Making Challenge (0.675), and Total CORGIS (0.881), all of which indicate adequate reliability [52].

In Experimental Design I, the means and standard deviations for the IMI and CORGIS scales and their respective subscales are presented based on the gameplay conditions (Figure 7). Except for the IMI Perceived Competence subscale, the first group playing under dynamic difficulty exhibited higher values across all IMI and CORGIS subscales compared to the second group playing under the normal difficulty condition.

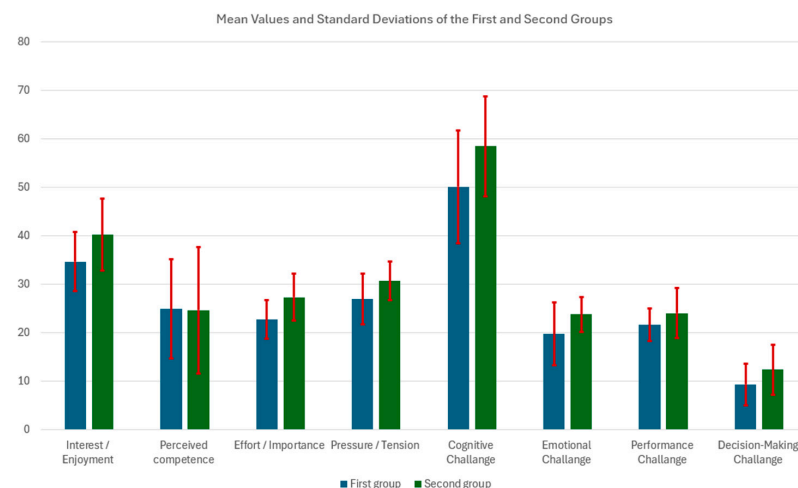


Figure 7. Graph of Means and Standard Deviations for CORGIS and IMI Subscales in Experimental Design I.

3.4. Participants

A total of 40 volunteer participants were included in the study. Among them, 11 were female and 29 were male, with a calculated mean age of 27.65 years. The age distribution

was as follows: 3 participants aged between 16 and 19, 9 participants between 20 and 24, 21 participants between 25 and 29, 5 participants between 30 and 40, and 2 participants between 40 and 60.

In the study, participants in the first group had a mean age of 27 years, whereas participants in the second group had a mean age of 28.3 years. The first group consisted of 5 women and 15 men, whereas the second group comprised 6 women and 14 men. The mean age of female participants in the first group was 30.6 years, while that of male participants was 25.8 years. In the second group, the mean age of female participants was 28.16 years, whereas that of male participants was 28.35 years. Despite random assignment of participants, the mean ages and gender distribution of the two groups were found to be comparable.

Participants reported an average of 15.83 years of video gaming experience, with a standard deviation of 6.35. This finding indicates that the sample predominantly comprised individuals with extensive gaming experience. In the study, participants in the first group reported a gaming experience of 14.35 years on average, whereas participants in the second group reported 17.3 years. It should be noted that these durations refer to the time elapsed since participants first started playing games and may not correspond directly to their current gaming frequency.

Participants were also asked about their preferred game genres. The distribution of preferences was as follows: Action ($n = 9$), Competitive ($n = 7$), Simulation ($n = 7$), Casual ($n = 5$), Role-Playing Games ($n = 5$), Adventure ($n = 3$), Strategy ($n = 2$), Story-Based ($n = 1$), and Massively Multiplayer Online (MMO) games ($n = 1$). The favorite genres of the participants in the first group were distributed as follows: Action ($n = 6$), Competitive ($n = 5$), Casual ($n = 3$), Simulation ($n = 2$), Role-Playing Games ($n = 2$), Story-Rich ($n = 1$), and Strategy ($n = 1$). The favorite genres of the participants in second group were distributed as follows: Simulation ($n = 4$), Action ($n = 3$), Adventure ($n = 3$), Role-Playing Games ($n = 2$), Competitive ($n = 2$), Casual ($n = 2$), Simulation ($n = 1$), MMO games ($n = 1$), Role-Playing Games ($n = 1$), and Strategy ($n = 1$). This distribution reflects that participants possessed diverse gaming experiences across different genres, contributing to a variety of perceptions toward the game.

3.5. Procedure

Participants were randomly assigned into two groups of 20. The first group played the game with the biofeedback-supported DDA system enabled. The second group first experienced the game with the DDA system disabled, then replayed it with the DDA activated.

Prior to participation, all participants were briefed on the study's content and provided informed consent for voluntary participation. General information regarding the functioning principles of the DDA system was presented, specifically noting that heart rate, voice level, and stress levels could influence in-game difficulty when the system was active.

The experiment was conducted on participants' personal computers using their existing hardware setups. Allowing all participants to use their own equipment eliminates the need for them to adapt to different setups and thus reduces variability arising from such adaptation. All sessions took place under standardized conditions in a dark, quiet room with stable lighting.

Participants wore the same smartwatch, provided and controlled by the researcher, but were not informed about the difficulty level they would play. This approach was adopted to prevent expectation effects and biases [53]. After synchronization between the smartwatch and the game was completed, participants commenced their gaming session.

The gameplay duration was limited to an average of 30 min. Participants were instructed to terminate the session if this time limit was exceeded. Following each session, participants completed a questionnaire that took approximately five minutes.

After the first session, all participants in the second group were given a short ten-minute break, during which their heart rate was monitored to confirm a return to baseline, and their subjective condition was also assessed through self-report. Subsequently, they replayed the game with the DDA system activated. At the end of the session, the same questionnaire was administered, and the data collection process was completed.

During gameplay, participants' physiological data and in-game interactions were recorded in real time using a monitoring and logging tool integrated into the game system.

In the first group (dynamic difficulty), 6 participants successfully completed the game, and 2 participants reached the time limit. In the second group (fixed difficulty), 8 participants completed the game, and 3 participants reached the time limit. When participants from the second group replayed the game under dynamic difficulty, 13 participants successfully completed the game.

3.6. Data Analysis

The Shapiro–Wilk test was applied to assess whether the data followed a normal distribution. The Shapiro–Wilk test is recommended when group sizes are less than 50 [52]. Although it is considered the most powerful test for samples under 50, it remains reliable for sample sizes up to 2000 [54]. For the comparison of two independent groups, normality and homogeneity of variance were verified, and consequently, the independent samples t-test was employed. This test is a suitable parametric method for analyzing mean differences between groups and yields robust results when assumptions are met [55]. For related measurements, where the same participants' scores under two different conditions were compared, the paired samples t-test was applied. This test is appropriate for determining whether differences between paired measurements are statistically significant [55]. The normality assumption for difference scores was verified using the Shapiro–Wilk test prior to analysis. In cases where data did not exhibit normal distribution and parametric test assumptions were not met, the Mann–Whitney U test was used to compare independent groups, while the Wilcoxon Signed Ranks Test was used for related measures. These non-parametric tests evaluate group differences based on ranked data. A statistical significance level of $p \leq 0.05$ was set, while $p \leq 0.01$ was considered highly significant.

In the first experimental design, IMI and CORGIS data were compared, and mean values along with standard deviations are presented in Table 3. The results from the Mann–Whitney U test are also shown, including Z, U, and p values.

In the Interest/Enjoyment subscale of the Intrinsic Motivation Inventory (IMI), the first group's average score ($\bar{x} = 40.25 \pm 7.383$) was significantly higher than that of the second group ($\bar{x} = 34.70 \pm 6.053$) ($U = 108.0$, $Z = -2.492$, $p \leq 0.05$, $r = -0.39$).

No significant difference was found between the first ($\bar{x} = 24.65 \pm 12.966$) and second groups ($\bar{x} = 24.95 \pm 10.241$) in the Perceived Competence subscale ($p = 0.968$).

In the Effort/Importance subscale, the first group scored significantly higher ($\bar{x} = 27.35 \pm 4.826$) than the second group ($\bar{x} = 22.75 \pm 4.024$) ($U = 88.5$, $Z = -3.030$, $p \leq 0.01$, $r = -0.48$).

For the Pressure/Tension subscale, the first group ($\bar{x} = 30.70 \pm 4.001$) also scored significantly higher than the second group ($\bar{x} = 26.95 \pm 5.195$) ($U = 108.5$, $Z = -2.488$, $p \leq 0.05$, $r = -0.39$).

The Total Intrinsic Motivation score was significantly higher in the first group ($\bar{x} = 122.95 \pm 16.916$) compared to the second group ($\bar{x} = 109.35 \pm 14.886$) ($U = 105.0$, $Z = -2.574$, $p \leq 0.01$, $r = -0.41$). This finding supports H1₁, while H0₁ is rejected.

Table 3. Mann–Whitney U Test Results Comparing IMI and CORGIS Mean Scores Between Players in Normal vs. DDA-Based Game Conditions.

		Mean $\pm$ Standard Deviation		Z	U	Cohen's r	p	q
		Second Group (Normal)	First Group (DDA)					
IMI	Interest/Enjoyment	34.70 $\pm$ 6.053	40.25 $\pm$ 7.383	−2.492	108.0	−0.39	0.013	0.022
	Perceived competence	24.95 $\pm$ 10.241	24.65 $\pm$ 12.966	−0.041	198.5	−0.01	0.968	0.968
	Effort/Importance	22.75 $\pm$ 4.024	27.35 $\pm$ 4.826	−3.030	88.5	−0.48	0.002	0.007
	Pressure/Tension	26.95 $\pm$ 5.195	30.70 $\pm$ 4.001	−2.488	108.5	−0.39	0.013	0.022
	Total: Intrinsic Motivation	109.35 $\pm$ 14.886	122.95 $\pm$ 16.916	−2.574	105.0	−0.41	0.010	0.022
CORGIS	Cognitive Challenge	50.10 $\pm$ 11.611	58.45 $\pm$ 10.354	−2.221	118.0	−0.35	0.026	0.037
	Emotional Challenge	19.80 $\pm$ 6.444	23.80 $\pm$ 3.650	−3.560	69.0	−0.56	0.000	0.000
	Performative Challenge	21.65 $\pm$ 3.391	24.10 $\pm$ 5.159	−1.212	155.5	−0.19	0.226	0.251
	Decision-Making Challenge	9.40 $\pm$ 4.284	12.40 $\pm$ 5.123	−2.139	122.0	−0.34	0.032	0.040
	Total: Perceived Challenge	100.95 $\pm$ 15.879	118.75 $\pm$ 16.814	−3.167	83.0	−0.50	0.002	0.007

In the CORGIS scale:

The Cognitive Challenge subscale showed significantly higher scores in the first group ($\bar{x} = 58.45 \pm 10.354$) compared to the second group ($\bar{x} = 50.10 \pm 11.611$) ($U = 118.0$, $Z = -2.221$, $p \leq 0.05$, $r = -0.35$).

The Emotional Challenge subscale also indicated significantly higher scores for the first group ($\bar{x} = 23.80 \pm 3.650$) than the second group ($\bar{x} = 19.80 \pm 6.444$) ($U = 69.0$, $Z = -3.560$, $p \leq 0.01$, $r = -0.56$).

No significant difference was found in the Performative Challenge subscale between the first ($\bar{x} = 24.10 \pm 5.159$) and second groups ($\bar{x} = 21.65 \pm 3.391$) ($p = 0.226$).

In the Decision-Making Challenge subscale, the first group ($\bar{x} = 12.40 \pm 5.123$) scored significantly higher than the second group ($\bar{x} = 9.40 \pm 4.284$) ($U = 122.0$, $Z = -2.139$, $p \leq 0.05$, $r = -0.34$).

The Total Perceived Challenge score was significantly higher in the first group ($\bar{x} = 118.75 \pm 16.814$) compared to the second group ($\bar{x} = 100.95 \pm 15.879$) ($U = 83.0$, $Z = -3.167$, $p \leq 0.01$, $r = -0.50$).

Heart rate range in Experimental Design I was compared using the Independent Samples *t*-Test (Table 4).

Table 4. Comparison of heart rate range data in Experimental Design 1.

Difficulty Type	N	Mean $\pm$ Standard Deviation	sd	t	p	Cohen's d	95% Confidence Interval of the Difference	
							Lower	Upper
Dynamic	20	15.50 $\pm$ 5.216	38	−5.294	<0.001	1.674	−2.390	−0.942
Normal	20	28.15 $\pm$ 9.326						

The analysis showed that participants playing under the normal difficulty condition exhibited a significantly higher heart rate range, aligning with the Relax-to-Win principle and supporting H1₄, while rejecting H0₄.

In addition, heart rate range for Experimental Design II was compared using the Paired Samples *t*-Test. In the comparison, the Paired Samples *t*-Test indicated that this difference was not statistically significant ($p > 0.05$).

A Mann–Whitney U test was conducted between the first group and the second dynamic group, and no statistically significant differences were observed in terms of the IMI and CORGIS measures ($p > 0.05$).

For Experimental Design II, IMI and CORGIS scores were compared using the Wilcoxon Signed Ranks Test, with results summarized in Table 5.

Table 5. Wilcoxon Signed Ranks Test Results for IMI and CORGIS Mean Scores in Experimental Design II.

		Mean $\pm$ Standard Deviation		Z	Cohen's r	p	q
		Second Group (Normal)	Second Group (Dynamic)				
IMI	Interest/Enjoyment	34.70 $\pm$ 6.053	41.25 $\pm$ 5.646	−3.830	−0.86	0.000	0.000
	Perceived competence	24.95 $\pm$ 10.241	28.05 $\pm$ 10.980	−1.852	−0.41	0.064	0.064
	Effort/Importance	22.75 $\pm$ 4.024	26.55 $\pm$ 5.185	−3.447	−0.77	0.001	0.002
	Pressure/Tension	26.95 $\pm$ 5.195	30.95 $\pm$ 4.914	−3.943	−0.88	0.000	0.000
	Total: Intrinsic Motivation	109.35 $\pm$ 14.886	126.80 $\pm$ 17.285	−3.824	−0.86	0.000	0.000
CORGIS	Cognitive Challenge	50.10 $\pm$ 11.611	54.15 $\pm$ 13.448	−2.761	−0.62	0.006	0.008
	Emotional Challenge	19.80 $\pm$ 6.444	22.55 $\pm$ 4.248	−2.950	−0.66	0.003	0.005
	Performative Challenge	21.65 $\pm$ 3.391	24.50 $\pm$ 4.696	−2.805	−0.63	0.005	0.007
	Decision-Making Challenge	9.40 $\pm$ 4.284	7.70 $\pm$ 4.168	2.571	0.57	0.010	0.011
	Total: Perceived Challenge	100.95 $\pm$ 15.879	108.90 $\pm$ 16.929	−3.180	−0.71	0.001	0.002

The Wilcoxon test results for the second group (Table 5) indicate that most measurements under dynamic difficulty were higher than under normal difficulty. Significant increases were observed in the Interest/Enjoyment ($Z = -3.830$, $p = 0.000$, $r = -0.86$), Effort/Importance ($Z = -3.447$, $p = 0.001$, $r = -0.77$), Pressure/Tension ($Z = -3.943$, $p = 0.000$, $r = -0.88$), Cognitive Challenge ($Z = -2.761$, $p = 0.006$, $r = -0.62$), Emotional Challenge ($Z = -2.950$, $p = 0.003$, $r = -0.66$), and Performative Challenge ($Z = -2.805$, $p = 0.005$, $r = -0.63$) subscales. Depending on these subscales, Intrinsic Motivation scale ($Z = -3.824$, $p = 0.000$, $r = -0.86$) and Perceived Challenge scale ($Z = -3.180$, $p = 0.001$, $r = -0.71$) showed a significant increase. In contrast, the Decision-Making Challenge subscale showed a significant decrease ($Z = 2.571$, $p = 0.010$, $r = 0.57$), and Perceived Competence did not show a significant increase ($p = 0.064$). These findings support $H1_2$, while rejecting $H0_2$.

Mann–Whitney U test results revealed significant differences between the first and second groups in both intrinsic motivation and perceived challenge levels. However, the Wilcoxon Signed Ranks Test, which examined the same participants' scores before and after the DDA experience, demonstrated stronger effects, particularly in Total Intrinsic Motivation ($Z = -3.824$, $p < 0.001$, $r = -0.86$) and Total Perceived Challenge ($Z = -3.180$, $p = 0.001$, $r = -0.71$). Effect size calculations showed that the Wilcoxon test yielded higher effect sizes across all subscales except Perceived Competence. While the Mann–Whitney test showed negligible effects for Perceived Competence, the Wilcoxon test results were nearly significant. In the test conducted for Experimental Design II, the only subscale that showed a decrease was Decision-Making. This decrease is presumed to be due to the fact that participants who had already played the game once, regardless of their success, had at least some idea about how to proceed in the game. This familiarity may also have encouraged some participants to try to improve their previous performance, thereby potentially increasing intrinsic motivation. It is assumed that players who completed the game on their normal difficulty approached the dynamic difficulty with greater confidence and aimed to perform better, whereas those who failed to complete it initially proceeded more cautiously in their effort to finish the game. These findings indicate that the impact of

the DDA-based game was more pronounced not only compared to the control group but also within the same individuals over time, thereby supporting H1₃ and rejecting H0₃.

4. Discussion

In recent years, the significance of personalized experiences in game design has been increasingly recognized, particularly through DDA mechanisms that respond to player reactions. However, the real-time implementation of such systems using biofeedback data remains limited. DDA systems dynamically adjust a video game's difficulty by processing data obtained from the player in real time. Traditionally, DDA adjusts difficulty by easing the game when the player shows signs of stress and struggle, and increasing the difficulty when the player is calm, aiming for an optimal challenge level—a pattern established in foundational DDA studies. However, a reverse approach called the “Relax-to-Win” model shifts this paradigm by requiring players to maintain calmness to succeed. In this framework, the game becomes harder when players are stressed and easier when they remain calm.

In this study, a biofeedback-based DDA system was developed and integrated within the “Relax-to-Win” principle, dynamically adjusting in-game difficulty according to the player's physiological state. To simplify the hardware requirements of DDA systems and enhance accessibility, data collection was achieved solely via a smartwatch instead of complex sensor setups. The DDA system was tested in a custom-developed survival-horror game titled *Code: Terror*, chosen for its genre-specific dynamics that intensify stress, anxiety, and threat perception—ideal for evaluating psychophysiological responses.

Findings from the IMI and CORGIS scales demonstrated that this system significantly enhanced player experience. Participants in the DDA exhibited higher intrinsic motivation and perceived a more engaging experience than those in the fixed-difficulty group. Furthermore, heart rate range data analysis revealed that players actively sought to calm themselves when DDA was enabled, embodying the “Relax-to-Win” approach.

In the IMI scale, the Interest/Enjoyment subscale showed that participants found the experimental condition more engaging and enjoyable. There was no significant difference in the Perceived Competence subscale, indicating that perceived competence remained stable across conditions. However, the Effort/Importance subscale indicated that participants perceived the experimental condition as more significant, exerting greater effort. The Pressure/Tension subscale further confirmed that the experimental condition induced moderate tension, reflecting the emotional impacts of the experience. Overall, the total IMI score suggests that participants derived greater enjoyment from the DDA-enhanced experience, thereby boosting intrinsic motivation.

For the CORGIS scale, higher scores in the Cognitive Challenge subscale confirmed that the experimental condition demanded more mental effort. The Emotional Challenge subscale also reflected greater emotional strain in the experimental group. No significant difference was observed in the Performative Challenge, indicating comparable performance expectations across conditions. In the Decision-Making Challenge subscale, the experimental condition appeared to stimulate decision-making processes more intensely.

Among participants who experienced both difficulty modes sequentially, scores in the dynamic difficulty condition were generally higher across subscales, except for Decision-Making Challenge, where no significant difference was observed, likely due to familiarity from prior gameplay. Similarly, Perceived Competence remained unaffected, suggesting that competence perception is relatively stable regardless of difficulty conditions.

Upon examination of the findings and analyses, it is evident that the impact of the DDA-based game emerged more prominently not only in comparison to the different groups but also within the same individuals over time.

Heart rate range analysis further confirmed that players in the normal difficulty mode exhibited higher heart-rate ranges compared to those in the DDA mode, reinforcing the “Relax-to-Win” model’s efficacy in promoting relaxation and stress reduction during gameplay.

It should be considered that participants’ heart rates may vary, which could influence the differences observed between groups. To account for this variability, the heart rate range (minimum–maximum) was analyzed instead of absolute values, allowing relative changes during gameplay to be observed. In future studies, including baseline measurements could be considered to more clearly distinguish the effects of gameplay.

The findings align with prior research by Moschovitis and Denisova [22] and Nogueira et al. [23], who similarly highlighted the potential of biofeedback-based DDA systems to optimize player experience in horror games. However, this study uniquely demonstrates the feasibility of using a smartwatch for biofeedback, enhancing practicality and accessibility while reducing hardware complexity. Unlike traditional horror games that rely on jump scares and atmospheric pressure, survival-horror games like *Code: Terror* provide a deeply personal experience of vulnerability, resource management, and constant threat—elements that naturally align with the “Relax-to-Win” model. The physiological coupling between player states and game mechanics enhances not only immersion but also sensitivity to individual differences. While prior work by Andrew et al. [46] explored DDA in survival-horror, it employed affective feedback rather than biofeedback and implemented an approach contrary to the “Relax-to-Win” model.

Despite meaningful results with a 40-participant sample, broader and more demographically diverse samples are recommended for generalization. Future studies could further explore how experienced and novice players respond to biofeedback-based DDA. Additionally, assessing whether the induced tension was universally perceived as a “positive challenge” would benefit from a more nuanced analysis of individual differences, including prior horror game experience and sensory sensitivity. Employing smartwatches with more advanced sensors may also yield more precise measurements.

Although participants were informed about the DDA system’s presence, no details about gameplay mechanics were disclosed. The significant effects on perceived difficulty and motivation suggest that the observed impacts stem from gameplay itself rather than cognitive expectations. Nonetheless, future studies should consider placebo-controlled designs to isolate these effects further.

Ultimately, *Code: Terror* transformed the gaming session into an emotional regulation space, showcasing the potential of games not just as entertainment but as tools for enhancing cognitive-emotional awareness. This suggests that biofeedback-driven DDA systems hold promises beyond gaming, particularly in domains like anxiety management, attention training, and therapeutic gaming applications. Expanding research across different genres and player profiles could further validate the system’s scalability and benefit in promoting in-game self-awareness.

5. Conclusions

This study examined the effects of a biofeedback-driven DDA system, based on the “Relax-to-Win” principle, using only a smartwatch for physiological data collection instead of multi-sensor systems. The custom-designed survival-horror game provided an optimal environment to evaluate the system’s efficacy under conditions that amplify psychophysiological responses.

Findings from the IMI and CORGIS scales demonstrate that the DDA system not only increased players’ intrinsic motivation but also made the game more enjoyable and engaging, while significantly affecting cognitive and emotional challenges. Heart rate range

data revealed that players tended to remain calmer, embodying the “Relax-to-Win” model. The system’s impact was evident even within the same individuals across different difficulty experiences, confirming that DDA positively influences not just in-game performance but also psychological balance.

In the analysis conducted across two different groups, significant differences were observed in the Interest/Enjoyment (Normal: 34.70 ± 6.053 ; DDA: 40.25 ± 7.383 ; $p \leq 0.05$) and Effort/Importance (Normal: 22.75 ± 4.024 ; DDA: 27.35 ± 4.826 ; $p \leq 0.01$) subscales of the IMI scale. These findings indicate that the DDA system enhances players’ intrinsic motivation, making the game more enjoyable and engaging. Additionally, Pressure/Tension scores increased (Normal: 26.95 ± 5.20 ; DDA: 30.70 ± 4.001 ; $p \leq 0.05$), suggesting that players reported greater emotional involvement and focus on the game.

According to the CORGIS scale results, significant effects of the DDA system were also observed in the Cognitive Challenge (Normal: 50.10 ± 11.611 ; DDA: 58.45 ± 10.354 ; $p \leq 0.05$), Emotional Challenge (Normal: 19.80 ± 6.444 ; DDA: 23.80 ± 3.650 ; $p \leq 0.01$), and Decision-Making Challenge (Normal: 9.4 ± 4.284 ; DDA: 12.4 ± 5.123 ; $p \leq 0.05$) subscales.

In the within-subject analysis, significant differences were again found in the IMI Interest/Enjoyment (Normal: 34.70 ± 6.053 ; DDA: 41.25 ± 5.646 ; $p \leq 0.01$) and Effort/Importance (Normal: 22.75 ± 4.024 ; DDA: 26.55 ± 5.185 ; $p \leq 0.01$) subscales. These results reinforce that the DDA system effectively increases players’ intrinsic motivation and enhances their engagement with the game. The Pressure/Tension scores also increased (Normal: 26.95 ± 5.195 ; DDA: 30.95 ± 4.914 ; $p \leq 0.01$), reflecting heightened emotional activation and focus during gameplay. The CORGIS scale further revealed significant effects of the DDA system on Cognitive Challenge (Normal: 50.10 ± 11.611 ; DDA: 54.15 ± 13.448 ; $p \leq 0.01$), Emotional Challenge (Normal: 19.80 ± 6.444 ; DDA: 22.55 ± 4.248 ; $p \leq 0.01$), and Performative Challenge (Normal: 21.65 ± 3.391 ; DDA: 24.50 ± 4.696 ; $p \leq 0.01$).

Although the heart rate range did not show a statistically significant difference between the normal and dynamic gameplay experiences for the second group ($p > 0.05$), the DDA system actively and continuously adjusted in-game parameters and difficulty. This indicates that the difficulty was indeed modified during gameplay, even if the overall heart rate distribution remained stable. Therefore, despite the absence of pronounced physiological differences between sessions, participants were still exposed to dynamic difficulty changes. This finding supports the interpretation that the differences observed in the survey results reflect genuine variations in perceived gameplay experience rather than bias or external factors.

Heart rate data demonstrated that players tended to remain calmer while interacting with the system, suggesting their internalization of the “Relax-to-Win” model. The system’s impact was evident even within the same individuals when experiencing different difficulty levels, further confirming that the DDA system positively influences not only in-game performance but also psychological balance. The “Relax-to-Win” approach is likely to be effective in more specific game genres. For instance, it is most applicable to games designed around tension and action. Additionally, it is expected to work in genres such as racing or sports games. In games with consistently high tempo, relaxation may not align with the flow of gameplay, whereas in story-driven or very low-tempo games, the system’s effect may be minimal or even negligible. This can be simply explained by the fact that the approach is effective if game performance is inversely related to stress. In some genres, players may find this type of biofeedback challenging. To mitigate this, a more controlled system could be implemented. Depending on the game design, integrating the model may be complex.

The use of smartwatches for biofeedback can be readily applied to different types of video games. Different game genres can elicit distinct physiological responses, necessitating

the development of various algorithms. However, maintaining player engagement while incorporating biofeedback requires careful balancing, as intrusive or overly frequent feedback could disrupt gameplay. In the future, smartwatches could serve as an accessible hardware tool to create more interactive video games and reach a broader audience. This approach can be easily used by anyone without causing discomfort, unlocking human-interactive video gaming with minimal hardware requirements. Future work could explore genre-specific adaptations, calibration procedures, and integration strategies that seamlessly combine biofeedback with game mechanics.

Because each player's experience may vary, especially considering their gaming experience and familiarity with the genre, more effective results could be achieved if the participant group in future studies was narrowed down to those with a similar level of familiarity with the genre. Less experienced players may react more, while those with a high level of familiarity with the genre may react less. Alternatively, there may be players who react more despite being familiar with the genre, and players who react less despite being unfamiliar with the genre.

Overall, the findings support that the DDA system positively affects not only in-game performance but also psychological equilibrium and cognitive–emotional awareness. However, due to limitations such as sample size and individual variability, future research should aim to test the system with more diverse player profiles and advanced smartwatch sensor technologies. In conclusion, biofeedback-based DDA systems hold significant potential not only for personalizing gaming experiences but also for broader applications in stress management, attention regulation, and emotional awareness.

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